Green Home Score & Retrofit Recommendation Engine Mid-Submission Progress Report (M2)

TEAM 103

October 10, 2026

1. Current Approach and Changes Since M1

Our core architectural vision from M1 remains grounded in separating deterministic physics and financial calculations from generative AI explanations. However, through system prototyping and empirical validation on our synthetic dataset of 5,000 properties, we have implemented key refinements across the scoring, backend infrastructure, and state isolation layers.

1.1 Architectural Refinements

- **Decoupled FastAPI Service Layer:** Rather than relying on static scripts, we implemented an asynchronous FastAPI backend paired with Uvicorn. This serves deterministic endpoints for property baseline evaluations (`/api/assessment`), scenario simulations (`/api/what-if`), and interactive retrofit package evaluations.
- **Certified Emission Factor Binding:** Calculations strictly enforce the UK Department for Energy Security and Net Zero (DESNZ) / DEFRA emission factors:
  - Grid Electricity: 0.20705 $\text{kgCO}_2\text{e/kWh}$
  - Mains Gas: 0.18254 $\text{kgCO}_2\text{e/kWh}$

  Accounting for separate fuel shifts ensures that increased electricity consumption from heat pump operation (modeled at COP = 3.2) does not erroneously mask gross gas reductions.
- **In-Memory Session State & Ground-Truth Immutability:** To enable arbitrary what-if experimentation without corrupting baseline customer data, we introduced an in-memory session abstraction. Users can inspect, alter floor areas, or adjust baseline energy figures on the fly. The engine recalculates metrics dynamically while leaving the dataset unchanged on logout or termination.
- **Strict Customer Identity Lookup:** We added strict customer and property identifier validation ($C10001$, $P10001$), rejecting unauthorized or non-existent IDs with explicit HTTP 404 responses.

1.2 Dual-Score Paradigm

To prevent homeowners from conflating current building efficiency with hypothetical retrofit improvements, our platform implements an explicit Dual-Score Paradigm separating baseline performance from simulated potential. The platform now implements an explicit *Dual-Score Engine*:

$$S_{\text{current}} = \sum_{i=1}^{7} w_i s_i^{\text{baseline}}, \quad S_{\text{potential}} = S_{\text{current}} + \Delta S_{\text{upgrades}} \tag{1}$$

The dashboard displays the un-retrofitted baseline (S_{current}) along with the dynamically simulated potential score ($S_{\text{potential}}$) in real time as specific interventions are toggled.

2. Experiments Run So Far and Their Results

2.1 Energy & Carbon Interaction Experiments

We simulated individual and combined fabric-first sequencing packages (P1–P5) across our synthetic cohort to verify that interactions between thermal envelope improvements and heating electrification do not double-count savings.

- **Fabric-First Diminishing Returns:** Applying cavity wall and loft insulation prior to heat pump sizing reduced space heating load by 25%–32%, lowering the required electrical throughput of the heat pump accordingly.
- **Maximum Net Carbon Offset:** For poorly insulated detached properties ($S_{\text{current}} \leq 40$), the combined deep retrofit package (P5: Fabric + ASHP + 8-Panel Solar PV) reliably delivered annual net carbon reductions between 2.85 tCO_2e and 3.36 tCO_2e.

Table 1: Benchmark Carbon Reductions Across Intervention Packages

Intervention Package	Δ Gas (kWh)	Δ Elec (kWh)
P1: Fabric Only (Loft+Wall)	$-4,250$	0
P2: Solar PV (Capped 2.5 MWh)	0	$-2,500$
P3: Heat Pump Only (COP 3.2)	$-14,800$	$+4,625$
P4: Fabric First + Solar	$-4,250$	$-2,500$
P5: Deep Electrification	$-14,800$	$+2,125$

2.2 Predictive Valuation Model Validation

We trained and evaluated the LightGBM hedonic property valuation regressor using 5-fold cross-validation on stratified price tiers.

- **Baseline (Median Model):** RMSE = £42,310, MAE = £33,480, $R^2 = 0.00$.
- **LightGBM Regressor:** RMSE = £11,840, MAE = £8,920, $R^2 = 0.884$.

The model predictions translate to conservative valuation uplift ranges of +2.4% to +4.8% following comprehensive band improvement (e.g., EPC E to B), aligning with prevailing UK residential green-premium studies.

2.3 Generative AI Groundedness & Resilience

We evaluated the Google GenAI SDK integration under rigorous prompt boundary constraints to ensure the advisor strictly references computed session variables.

- **Deterministic Groundedness:** Across 50 structured benchmark prompts, the AI achieved 96.0% numeric alignment with backend calculations, completely eliminating fabricated baseline scores.
- **Domain Boundary Guardrails:** When tested with out-of-scope queries (e.g., adding multi-split air conditioners or private EV chargers), the model acknowledged baseline physical trade-offs (e.g., marginal cooling electrical load) while redirecting users to supported space heating and fabric options.
- **Transient Resilience:** To protect against Google API 503 Service Unavailable spikes, we built an automatic failover pool across `gemini-2.5-flash`, `gemini-1.5-flash`, and `gemini-2.5-flash-lite`, achieving 99.8% call completion across simulated high-load runs.

3. Progress on Testing

We have implemented a test suite consisting of 37 automated test cases using `pytest`. All 37 tests currently execute cleanly with zero assertions failing:

- **Physics & Carbon Engine (`test_engines.py` – 24 tests):**
  - Monotonic scoring compliance: Verifies that upgrading any single attribute never reduces the sub-score or final composite score.
  - Component weight consistency: Ensures exact $\sum w_i = 1.0$ partitioning.
  - DEFRA emission factor math: Verifies directional sign correctness for fuel substitutions.
  - Boundary condition tests: Confirms score capping at $[0, 100]$ and solar yield caps at $2,500$ kWh/year.
- **Session State & Authentication (`test_api.py` – 8 tests):**
  - Rejection of unauthorized customer IDs with HTTP 404.
  - Verification that what-if modifications remain ephemeral and do not overwrite synthetic CSV/JSON data files.
  - Float parsing validation, ensuring that non-integer attributes (e.g., floor area of 115.3 m^2) parse without truncation.
- **AI Advisor Engine (`test_llm.py` – 5 tests):**
  - Integration with Google GenAI SDK.
  - Fallback activation under simulated network failure.
  - Strict rejection of out-of-bounds calculations in prompt payloads.

4. Plan for the Remaining Time

To prepare the system for final submission and evaluation, we will prioritize:

1. **Frontend UI Polish:** Finalize the dual-score visual gauge, streamline mobile responsiveness, and polish the Retrofit Battle package comparison cards.
2. **Batch Evaluation Hardening:** Run batch stress testing across the remaining synthetic records to benchmark API latency and eliminate edge-case edge failures.
3. **Deliverable Packaging:** Freeze the repository branch on `origin/main`, clean documentation, and compile the final evaluation report.